

boot_android_emulator.sh

Field	Value
Revision	2
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§11.4.18	In-source doc block present in scripts/ boot_android_emulator.sh

Overview

Boots an Android AVD on a remote Linux host (`nezha.local`) through the `vasic-digital/containers` submodule (§11.4.76) — or with a clean direct fallback when the submodule’s Go tooling is not built for the target.

Prerequisites

- SSH key-based access to the remote host
- Android SDK + desired AVD installed on the remote host (see `EMULATED_DEVICE_TESTING.md`)
- `libbsd.so.0` on the remote host (emulator dependency)
- Containers submodule checked out at `containers/` (project root sibling)

Usage

```
# Default boot (CZ_API36_Phone on nezha.local)
```

```
bash scripts/boot_android_emulator.sh
```

```
# Custom AVD and host
```

```
SSH_HOST=other-host.local AVD=Pixel_9a bash scripts/  
boot_android_emulator.sh
```

```
# Custom port and resources
```

```
PORT=5556 RAM_MB=4096 CORES=4 bash scripts/  
boot_android_emulator.sh
```

Environment Variables

Variable	Default	Description
SSH_HOST	nezha.local	Remote host
SSH_USER	milosvasic	SSH user
AVD	CZ_API36_Phone	AVD name
PORT	5554	Emulator console port
ANDROID_SDK_ROOT	/home/milosvasic/ Android/Sdk	SDK path on remote
LD_LIBRARY_PATH_EXTRA	/home/ milosvasic/.local/ lib	Extra lib path (libbsd)
RAM_MB	3072	Emulator RAM
CORES	2	Emulator CPU cores
GPU_MODE	swiftshader_indirect	GPU acceleration
COLD_BOOT	true	Force cold boot (wipe)
BOOT_TIMEOUT_SEC	180	Boot timeout

Outputs

- **Evidence directory:** docs/qa/<run-id>-android-emu-boot/
- **Attestation:** attestation.json with device properties + boot metrics
- **State file:** emu_state.env for consumption by other scripts
- **SSH tunnel:** ADB available at localhost:<PORT>

Remote-process full-detachment hardening (§11.4.144)

The remote emulator launch on nezha is wrapped in **setsid nohup ... </dev/null >log 2>&1 &** for **true full detachment** from the launching SSH session. The plain nohup & used previously left the remote qemu/AVD process tied to the SSH session's process group: if that SSH session ended mid-boot-wait (operator interrupt, connection drop), the remote emulator exited gracefully and the final SSH-tunnel / attestation step was never reached — a §11.4.144 tracked-device-availability gap (the launched device silently went away with the session rather than being followed independently).

setsid puts the remote launch in its **own session + process group** so it survives the launching SSH session ending; </dev/null detaches stdin and >log 2>&1 redirects all output to a remote log file the next reconnect can read. The boot itself was already PROVEN (emulator-5554, API 36, boot_completed=1); this hardening ensures the boot SURVIVES an interrupted launch session so the attestation step is reachable on reconnect.

Honest boundary (§11.4.6): on-target *persistence-after-reboot* verification (re-boot + mid-boot interrupt test, which leaves remote state) is operator-attended and not asserted here — the `setsid` change addresses the SSH-session-detachment class; full persistence proof is a separate operator-attended item.

Cleanup

The script does NOT auto-kill the emulator (to allow multiple sessions to use the running instance). To stop:

```
ssh nezha.local "ps aux | grep qemu | grep -v grep | awk '{print \
    $2}' | xargs -r kill"
```

Or use the containers submodule's `emulator-cleanup` command when built.

Cross-references

- [docs/design/EMULATED_DEVICE_TESTING.md](#)
- [containers/pkg/emulator/](#)
- [containers/cmd/emulator-matrix/](#)
- [containers/images/android-test/](#)
- [scripts/boot_android_emulator.sh](#)
- §11.4.76 Containers-submodule mandate
- §11.4.161 Rootless container runtime mandate

Last verified

2026-06-22 — boot on nezha.local (Linux x86_64, KVM, Android API 36); §11.4.144 `setsid` full-detachment hardening added to the remote launch.